Same Strategy, Two Results: Eleven Causes of Divergence between Backtest and Live Trading

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Abstract

A backtest and a live trading bot can run the same strategy code and still trade differently. This note documents eleven such divergences found during 2026 in a multi-strategy gold trading bot on MetaTrader 5 and its event-driven backtest engine. They fall into four kinds: features that existed on only one side, the same feature implemented differently on each side, plumbing that exists only in live trading, and a backtest market model that did not match the broker. Several were large enough to reverse a strategy's result: a filter that improved the backtest did not run live at all, a break-even rule closed a trend trade at $45 instead of its $150 target, and a recovery routine opened a position nobody had asked for. The note closes with the trade-by-trade reconciliation that found most of them, and with a week in which live and backtest agreed to within 0.1 R.

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1 Why this note

A backtest answers the question "what would this strategy have done", but only for the strategy as the backtest engine understands it. In a system where the backtest engine and the live execution layer are separate code, every rule a strategy relies on, from a filter to a stop adjustment, has to be implemented twice, and every place where the two implementations differ is a place where live results will drift from the backtest without any error being raised. A companion note covers errors in the data and in the backtest itself; this one covers what appears only when the two sides are compared.

Exhibit 1. The eleven divergences

#	Divergence	How it was found	Effect in live trading
Feature on one side only			
1	Regime and trend filters applied only by the backtest engine	configuration traced into live code	filters never ran live; the backtest version also looked ahead
2	Order-flow filter read a historical file live	filter value checked live	filter silently inactive
3	Trailing-stop settings ignored by live trading	exit path traced	a tuned exit existed only in the backtest
Same feature, different implementation			
4	Grid batch closed as a group in the backtest, position by position live	same day compared batch by batch	profitable batches closed at a loss; new batches opened
5	Break-even settings of the signal ignored; lock level ahead of price	a live trade closed too early	$45 instead of $150; strategy's sign reversed
6	Position-size multiplier ignored live	question about sizing traced in code	positions 67% larger than in the backtest
Live-only plumbing			
7	Recovery after restart placed a take-profit as a limit order	unexplained position on the account	an unwanted long position opened
8	Deal lookup used local time instead of server time	warning on every order	stop could be placed on another position
9	Contract size missing from configuration	order rejected by the broker	order 100 times too large
Backtest market model			
10	Short stops checked on bid instead of ask	live trades reconciled one by one	9 missed stops in 972 shorts
11	Trend filter silently off until warmed up	live trades reconciled one by one	23 backtest entries against 5 live

2 Features on one side only

1. Filters that only the backtest applied

What happened. Regime filters for the grid strategies and a trend filter for a dip-buying strategy were written as blocks in the strategy configuration and applied by the backtest engine before signals reached execution. Live trading never read those blocks. Every improvement those filters brought to the backtest was an improvement that did not exist in live trading.
How it was found. Following one configuration key from the file into the code: it appeared in the backtest engine and nowhere in the live path.
Effect. The filters were rebuilt inside the strategies themselves, so that both sides run the same code. That exposed a second problem: the engine's version computed the four-hour state with a resample that included the still-forming bar, using minutes that had not happened yet. With the honest version, the dip-buying strategy's portfolio test fell from $146,000 to $88,000. The lower number is the real one.

2. A live filter reading yesterday's file

What happened. A filter based on CME gold order flow was validated out of sample and switched on for a trend strategy. Its data source was the historical file used by the backtest. In live trading that file ended in the past, the filter received no value for the current minute, and a missing value meant "do not block".
How it was found. Checking what value the filter actually saw in live trading, rather than whether it was enabled.
Effect. The filter had been switched on but had never acted. A live data service was built, and the live and historical calculations were compared minute by minute over 2,758 minutes: the largest difference was 0.00005 and the sign agreed every time. The live path now has no fallback to the file: a missing live value is reported, not silently replaced by old numbers.

3. An exit rule that live trading did not read

What happened. A breakout strategy passed its trailing-stop settings to execution through the signal's metadata. The backtest read them; live trading used its own trailing system and ignored them. A fix that improved the strategy's exit, worth +$438 against −$95 over a year in the backtest, could only ever change the backtest.
How it was found. While testing the fix, by tracing what the live exit path would do with the new setting.
Effect. The fix was kept out of live trading until the exit paths are unified. The gap had existed before the fix as well; the fix only made it visible.

3 The same feature, implemented differently

4. A grid batch closed as one in the backtest and piece by piece live

What happened. Grid strategies open several positions as a batch and close them together when the batch's combined profit falls back from its peak. The backtest closed all positions of a batch in one transaction at one price. Live trading instead moved each position's stop on the broker's server, and the broker closed them one at a time.
How it was found. On one day the live and backtest signals matched 28 of 28, but the results did not. Comparing a single batch side by side showed the mechanism (Exhibit 2).
Effect. A batch that closed with a profit in the backtest closed with a loss live. Worse, positions closing separately freed their grid levels, so the strategy opened new batches where the backtest had one: three short batches live in place of one batch held for over two hours. The broker-side stops were switched off for parity, at the cost of relying on a wide emergency stop if the bot crashes.

Exhibit 2. One grid batch, 14 April 2026: backtest against live

	Backtest		Live	
Position	Exit	PnL, $	Exit	PnL, $
1 (entry about 4,786)	4,785.08	+0.80	4,787.35	−5.00
2 (entry about 4,788.5)	4,785.08	+3.60	4,787.35	+4.00
Batch		+4.40		−1.00

Notes:
– Backtest: both positions closed together when the batch's profit fell back from a peak of 0.85%.
– Live: the broker closed each position at its own stop, two seconds apart, after the batch peaked at 0.99%; the first closed at a loss.
– Both positions were shorts.
Source: bot trade log and backtest trade log.

5. A break-even rule that ignored the strategy

What happened. Each signal says whether its position should move the stop to break-even, and when. The backtest read those settings. Live trading registered every position with a general rule instead: move the stop once the trade is 1% in profit. Strategies that had switched break-even off were given it anyway. For some trades the rule was also wrong in itself: it locked half of the distance to the target, which for a short could put the new stop on the profitable side of the current price, so the position closed at once.
How it was found. A short on the live account closed at break-even, $44.65 in profit, one second after break-even activated, in a trend strategy whose target was $150.
Effect. For a strategy with a 22% win rate the winners must reach their targets. This had already been seen in the backtest, when an early break-even was tried as a way to reduce the strategy's risk: it cut the winners and turned the result negative. Over the seven weeks before the live trade the backtest had 18 trades; four winners at $150 and fourteen losers at $20 make +$320, while the same four winners cut to $44.65 make −$101. The strategy therefore runs without break-even, and live trading had applied it regardless. Live trading now uses the signal's own settings and delays activation until the locked level is behind the price.

6. A size multiplier that live trading did not read

What happened. Grid strategies size each level of a batch at a fraction of the normal risk, 60% in one of them, passed as a multiplier in the signal. The backtest applied it. Live trading sized every level at the full risk.
How it was found. Answering a question about whether risk is set per position or per batch, by tracing the multiplier through the code: it appeared in the backtest engine and nowhere in live execution.
Effect. Live positions in that strategy were about 67% larger than in the backtest, and larger still at the levels where an extra depth scaling applied. Results matched in direction but not in size, which is easy to mistake for ordinary slippage.

4 Plumbing that exists only in live trading

7. A take-profit placed as a limit order

What happened. After a restart, a recovery routine checks that every open position has its stop and target at the broker. It did not recognise a target set on the position itself and placed a new one as a buy limit order. On MetaTrader 5 a limit order does not close a position; it opens a new one. When the price reached the level, the short closed at its own target and the limit order opened a long that the bot did not track.
How it was found. An unexplained 0.01-lot long without a stop appeared on the demo account.
Effect. A second fault in the same routine moved stops to break-even at restart: it computed profit as a percentage 100 times too large and confused buy and sell with long and short. A gold short 0.33% in profit was given a stop $0.70 from the price and closed three minutes later. Targets are now set on the position itself, and break-even at restart follows the same rules as before the restart.

8. Looking for a trade in the wrong time zone

What happened. After sending an order, the bot looks up the resulting deal to find the position's ticket. The query used the local clock, while the broker records deals in server time two or three hours ahead, so the ten-second search window never contained the deal. A fallback then took "the newest position" as the ticket.
How it was found. The warning "deal not found, using newest-by-time fallback" appeared on every order, 9 of 9.
Effect. With several positions opening at once, as in a grid batch, the fallback can hand the same ticket to all of them, and the stop meant for one position is placed on another. A test with three simultaneous orders on a demo account confirmed it: the new lookup by ticket returned three correct tickets, the old path would have returned the same ticket three times.

9. A contract size of one ounce

What happened. Order size is computed from the notional value divided by the price and the contract size. A broker symbol missing from the instrument configuration received the default contract size of one; a gold lot is 100 ounces.
How it was found. The broker rejected an order for 44.44 lots on a $6,200 demo account for insufficient margin; the intended size was 0.44 lots.
Effect. A rejection is the lucky outcome. With a large enough balance or leverage the order would have been filled at 100 times the intended risk.

5 The backtest's model of the market

10. Short stops checked on the bid

What happened. Bars are built from bid prices. The engine checked whether a short position's stop or target had been hit using the bid high and low, while the broker closes a short at the ask.
How it was found. Live trades were reconciled with the backtest one by one. A short was stopped live when the ask reached 4,300.38 against a stop at 4,300.30; the bid never went above 4,300.25, so the backtest kept the trade open and later closed it at break-even.

Effect. An audit of 972 shorts found 9 missed stops and 4 false take-profits. Adding the spread to the bar's high and low for shorts reduced the portfolio result by 0.9% and one short-only strategy's by 7%.

11. A filter that is silently off until it has warmed up

What happened. A trend filter on four-hour bars needs 54 closed bars, about nine trading days, before it can decide. In the backtest it let every trade through until then, without logging a rejection, so a short backtest traded without the filter that live trading applied.
How it was found. A 15-day backtest, run to compare with live trading, made 23 entries over 10–18 September against 5 live. Recalculating the filter by hand showed that 10 of the 11 disputed entries would have been blocked.
Effect. The short backtest lost $151 where live trading lost $551. Backtests used for comparison with live trading now start three to four weeks before the period of interest, and the presence of each filter's rejection counter is checked.

6 How the divergences were found

Most of the eleven were found by one procedure: reconcile every live trade with the backtest of the same days, one trade at a time.

- **Match by ticket, not by time.** The broker's position report gives tickets in server time; the bot's log links each ticket to the strategy and signal that produced it; the backtest's trade log gives the trades it would have made. Two strategies trading in the same minute are separated by side and order of signals.
- **Start the backtest early.** The backtest must start weeks before the live period so that every filter and indicator is in the same state as the live bot's (cause 11).
- **Look at each unmatched trade.** A trade present on one side only is the most informative observation: it found causes 4, 10 and 11, and the unexplained position behind cause 7.
- **Trace each configuration setting into both code paths.** A setting that appears in the backtest engine and not in live execution, or the reverse, is a divergence waiting for a trade to reveal it (causes 1, 3 and 6).
- **Check what a filter saw, not whether it is enabled** (cause 2).

When the two sides do agree, the comparison is precise. Exhibit 3 shows the first live week of a strategy that shorts gold into the London morning fix: eight trades, every one matched to the backtest on the same day and leg, entry prices within $0.01–0.34, and a total of -0.69 R live against -0.61 R in the backtest.

Exhibit 3. A week in which live and backtest agreed, 28 September – 5 October 2026

Date	Leg	Live entry	Backtest entry	Live, R	Backtest, R
28 Sep	Main short	4147.74	4147.50	+0.55	+0.49
28 Sep	Rebound long	4145.61	4145.46	+0.89	+0.90
29 Sep	Main short	4141.13	4141.07	−0.03	−0.04
30 Sep	Main short	4185.16	4185.05	−0.75	−0.75
1 Oct	Main short	4160.99	4160.93	−0.08	−0.10
2 Oct	Main short	4185.10	4185.05	−0.80	−0.78
5 Oct	Main short	4167.76	4168.10	+0.56	+0.69
5 Oct	Rebound long	4165.08	4165.09	−1.01	−1.02
Total				−0.69	−0.61

Notes:
– Live: MetaTrader 5 Expert Advisor on a live account; backtest: the same strategy settings in the backtest engine.
– Results in R, the loss at the stop, after spread and commission, so that different position sizes are comparable.
Source: broker position report and backtest trade log.

7 Conclusion

The eleven divergences share one cause: two implementations of the same intent. Features added to the backtest engine for convenience, settings passed through metadata that only one side reads, and live machinery that the backtest never exercises all let the two sides drift apart without an error. The remedies are structural, not statistical: keep strategy rules inside the strategy so that both sides run the same code, make missing live inputs loud instead of silently permissive, and reconcile live trading with the backtest trade by trade from the first week, while the trades are few enough to examine one at a time.